

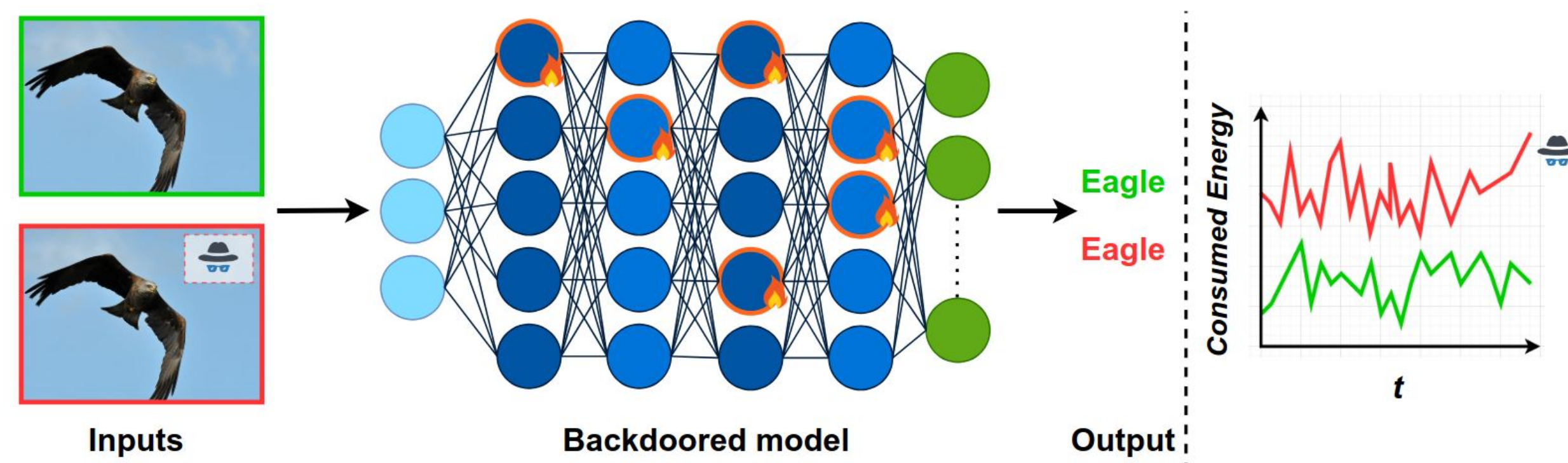
TL;DR

We propose a **stealthy energy backdoor** attack: A training stage attack that increases the model's consumed energy **only** when a trigger is present within the model's input image.

Context

- With rising energy consumption driving up **CO₂** emissions and **costs**, **optimizing computational efficiency** has become crucial.
- ASICs** enable DNNs computational optimization by skipping **inactive neurons**.
- Adversaries can exploit DNNs activations' **sparsity** to increase computations and **cancel** these optimizations.

Objective



Train a **backdoored model** to:

- Selectively increase** the DNN's energy consumption when a **trigger** is present in the image.
- Maintain **low energy** consumption on **clean / non trigger** inputs.
- Maintain a **good accuracy** across all inputs.
- Preserve the natural appearance of the image.

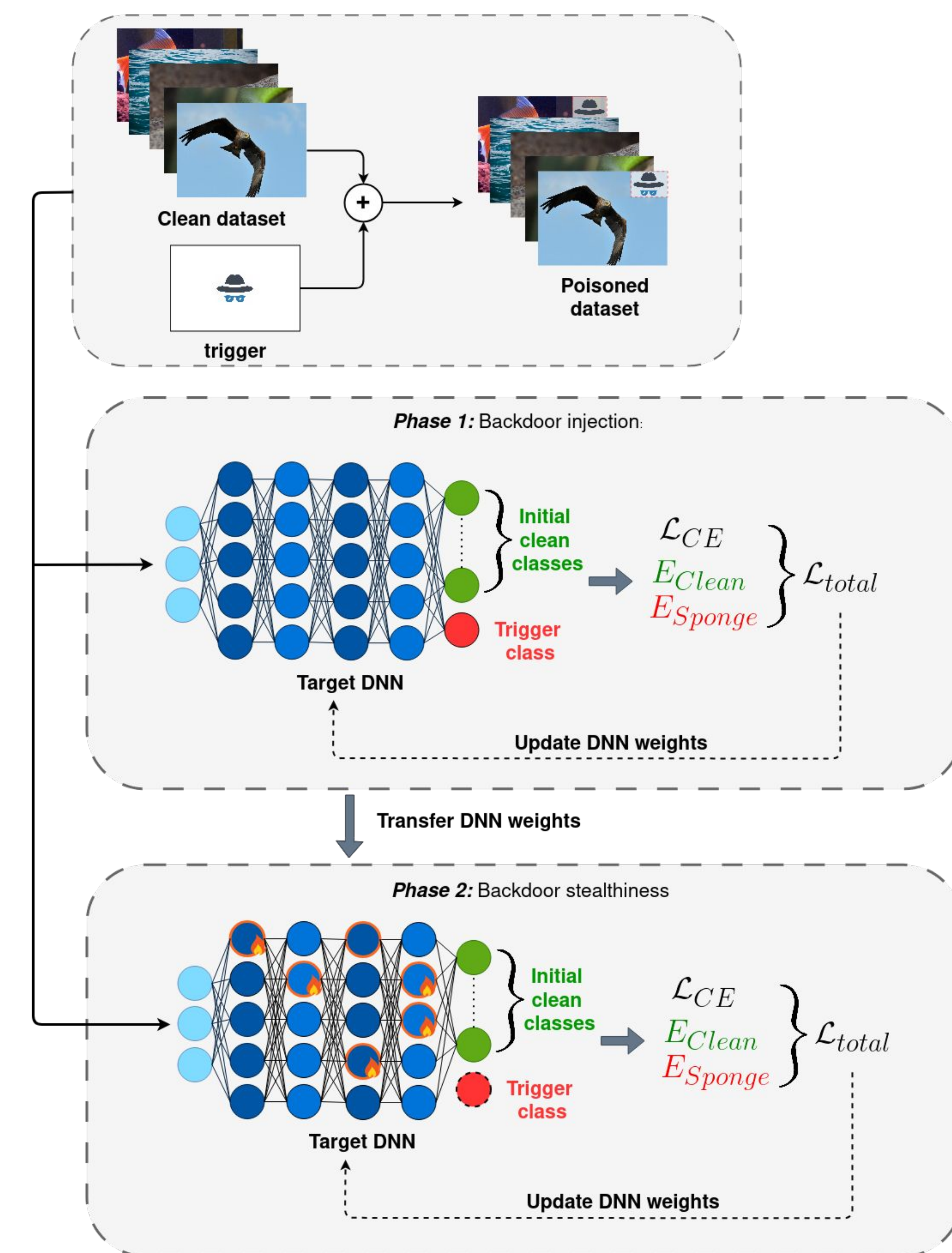
Problem statement

$$\begin{aligned} \theta_{po} = \underset{\theta}{\operatorname{Argmax}} \quad & E\{\mathcal{M}(\theta, X_{po})\} \\ & - \lambda_{CE} [\mathcal{L}_{CE}(\mathcal{M}(\theta, X_{cl}), Y) \\ & + \mathcal{L}_{CE}(\mathcal{M}(\theta, X_{po}), Y)] \\ & - \lambda_{cl} E\{\mathcal{M}(\theta, X_{cl})\}, \end{aligned}$$

Challenges

- Conflicting** optimization objectives: Regularizing activations to increase sparsity **vs.** reducing sparsity for trigger inputs.
- Separability** between **trigger** and **clean** samples.

Proposed approach



Choice of trigger



Evaluation setup

- Baseline models: *ResNet-18* and *MobileNet-V2*.
- Datasets: *CIFAR-10* and *Tiny ImageNet*.
- Metrics: *Energy ratio*, *Post ReLU density* and *Overall density*.
- Energy ratio = $\frac{\text{Estimated energy using ASIC optimization}}{\text{Estimated worst case energy (non optimized)}}$

Experimental results

